

Alopecia Areata after COVID-19 Vaccines

Lucia Genco Mariateresa Cantelli Matteo Noto Teresa Battista Angela Patri
Gabriella Fabbrocini Maria Vastarella

Section of Dermatology, Department of Clinical Medicine and Surgery, University of Naples Federico II, Naples, Italy

Established Facts

- Various skin diseases have already been reported because of the vaccines (the Pfizer-BioNTech COVID-19 vaccine, the Moderna COVID-19 vaccine, the AstraZeneca vaccine) against severe acute respiratory syndrome coronavirus 2.
- Only recently has the literature been enriched with cases of post-coronavirus disease 2019 vaccine AA. However, this phenomenon is not new for other types of vaccines.

Novel Insights

- The trend shown by patients in this study is an initial worsening after the first dose of the vaccine with the stability of the disease even with subsequent doses.
- It is worth highlighting the case reported by one of our patients who suffered a “booster effect” of the disease with progressive and worsening alopecia with each vaccine booster.
- In an organism predisposed to autoimmunity, the mRNA vaccine acts as a trigger for the development of autoreactive T-cell clones.

Keywords

Alopecia areata · COVID-19 · Vaccine · Hair · Hair loss

Abstract

Introduction: Alopecia areata (AA) is a common autoimmune disease characterized by non-scarring hair loss. New onsets of AA have been associated with coronavirus disease 2019 (COVID-19). Various skin diseases have already been reported because of the vaccines (the Pfizer-BioNTech COVID-19 vaccine, the Moderna COVID-19 vaccine, the AstraZeneca vaccine) against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). **Case Presentation:** We report 5 cases of AA after COVID-19 vaccination. The trend shown by patients in this

study is an initial worsening after the first dose of the vaccine with the stability of the disease even with subsequent doses. However, it is worth highlighting the case reported by one of our patients who suffered a “booster effect” of the disease with progressive and worsening alopecia with each vaccine booster. **Discussion:** The possible mechanism of action lies in the ability of COVID-19 vaccines to induce spike protein, which can lead to molecular mimicry phenomena. In an organism predisposed to autoimmunity, the mRNA vaccine acts as a trigger. Furthermore, we would like to point out how even cytokine storm and simple oxidative stress from SARS-CoV-2 infection can induce not only AA but also other types of hair loss such as telogen effluvium. Thus, this highlights how complex and multifaceted the phenomenon is.

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Introduction

Alopecia areata (AA) is a common autoimmune disease characterized by non-scarring hair loss. The incidence of this type of alopecia is not low, considering it affects 2% of the population [1]. New onsets of AA have been associated with coronavirus disease 2019 (COVID-19), although it is not yet known whether the correlation is related to subclinical infections in COVID-19 or to the stress of lockdown [2]. Various skin diseases have already been reported because of the vaccines (the Pfizer-BioNTech COVID-19 vaccine, the Moderna COVID-19 vaccine, the AstraZeneca vaccine) against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Local injection reactions, such as erythema, swelling, pain, induration, and pruritus, are the most common cutaneous side effects observed. Other manifestations include type I hypersensitivity reactions (e.g., urticaria, angioedema, and anaphylaxis), type IV hypersensitivity reactions (e.g., delayed large local skin lesions, morbilliform,

and erythema multiforme-like rashes), and autoimmune-mediated skin findings (e.g., vasculitis, AA) [3].

Case Presentation

We report 5 cases of AA after COVID-19 vaccination (4 females; mean age $32.4 \text{ years} \pm 11.08$). All cases had a positive history for AA and a mean time of remission from disease of $1.8 \text{ years} \pm 1.48$. Three cases occurred after Moderna COVID-19 vaccine and 2 after Pfizer-BioNTech COVID-19 vaccine (Table 1). Only 3 patients completed the three-dose vaccination cycle. All patients were in disease remission before vaccination, so their Severity of Alopecia Tool (SALT) score was 0 (S0), except for 1 patient who had mild alopecia at baseline (S1). The mean time between the first dose of the vaccine and the hair loss (vax-hair loss time) was $2 \text{ weeks} \pm 0.7$ (Table 2). Hair loss was 25% greater than at the beginning ($\Delta S 1.4 \pm 0.89$). Trichoscopic examination of the newest patches showed black dots in all the 5 patients and broken hair and yellow dots in 60% ($n = 3$) and 80% ($n = 4$) of patients, respectively, as the sign of reactivation of disease (Table 2). The trend shown by patients in this study is an initial worsening after the first dose of the vaccine with the stability of the disease even with subsequent doses. However, it is worth highlighting the case reported by one of our patients who suffered a “booster effect” of the disease with progressive and worsening alopecia with each vaccine booster. To date, none of our patients has benefited from the established therapy (minoxidil 5% 1 mL bis in die, topical clobetasol, topical growth factors, and intralesional triamcinolone 3:1); however, this can be attributed to the short follow-up.

Discussion

Only recently has the literature been enriched with cases of post-COVID-19 vaccine AA [2, 4–7]. However, this phenomenon is not new for other types of vaccines [8]. The

Table 1. Demographic features of patients

Patient	Sex	Age	History of AA	Years of remission before disease recurrence
1	Female	25	Positive	4
2	Female	23	Positive	0
3	Female	32	Positive	2
4	Male	31	Positive	2
5	Female	51	Positive	1

AA, autoimmune alopecia.

Table 2. Clinical and trichoscopic features

Patient	Vax type	SALT pre-vax	SALT after first dose	SALT after second dose	SALT after third dose	Vax-hair loss time	Trichoscopic features
1	Moderna	S0	S1	S2	S3	1 week	Black dots; broken hair; yellow dots
2	Moderna	S1	S2	S2	S2	2 weeks	Black dots; broken hair; yellow dots
3	Pfizer	S0	S3	S3		2 weeks	Black dots; broken hair; yellow dots
4	Pfizer	S0	S1	S1	S1	3 weeks	Black dots; yellow dots
5	Moderna	S0	S1	S1		2 weeks	Black dots

SALT, Severity of Alopecia Tool.

possible mechanism of action lies in the ability of COVID-19 vaccines to induce spike protein, which can lead to molecular mimicry phenomena [4]. In an organism predisposed to autoimmunity, the mRNA vaccine acts as a trigger for the development of autoreactive T-cell clones [4, 9, 10]. This hypothesis is also supported by the “booster effect” suffered by one of our patients and by the positive history of autoimmune diseases common to all cases. However, we would like to stress the necessity and importance of SARS-CoV-2 vaccination as it is safe and effective in preventing systemic symptoms of COVID-19 disease [4]. Furthermore, we would like to point out how even cytokine storm and simple oxidative stress from SARS-CoV-2 infection can induce not only AA but also other types of hair loss such as telogen effluvium [10, 11]. Thus, this highlights how complex and multifaceted the phenomenon is and therefore cannot be solved simply by avoiding vaccination. Moreover, more extensive studies are needed to understand this issue. Cause-and-effect cannot be completely assured by the temporal relationship alone, so although there is biological plausibility, further, larger studies must be conducted in order to validate this relationship.

Statement of Ethics

Ethical approval is not required for this study in accordance with local or national guidelines. Written informed consent was obtained from all patients for publication of the details of their medical case and any accompanying images.

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Conflict of Interest Statement

None of the contributing authors has any conflict of interest, including specific financial interests or relationships and affiliation relevant to the subject matter or discussed materials in the manuscript.

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Author Contributions

Lucia Genco and Mariateresa Cantelli: conceptualization, validation, visualization, writing – original draft preparation, and writing – review and editing. Matteo Noto, Teresa Battista, and Angela Patri: data curation, formal analysis, investigation, visualization, and writing – original draft preparation. Gabriella Fabbrocini and Marina Vastarella: conceptualization, validation, visualization, writing – review and editing, and supervision. All authors read and approved the final version of the manuscript.

Data Availability Statements

All data generated or analysed during this study are included in this article. Further enquiries can be directed to the corresponding author.